

SDoF V1 Fastener Guide

Fall 2023

Common Fasteners used on SDoF*



1/4-20 button head,
0.625" long

Part number:
MCMaster-92949A539



Assorted M3
machine screws

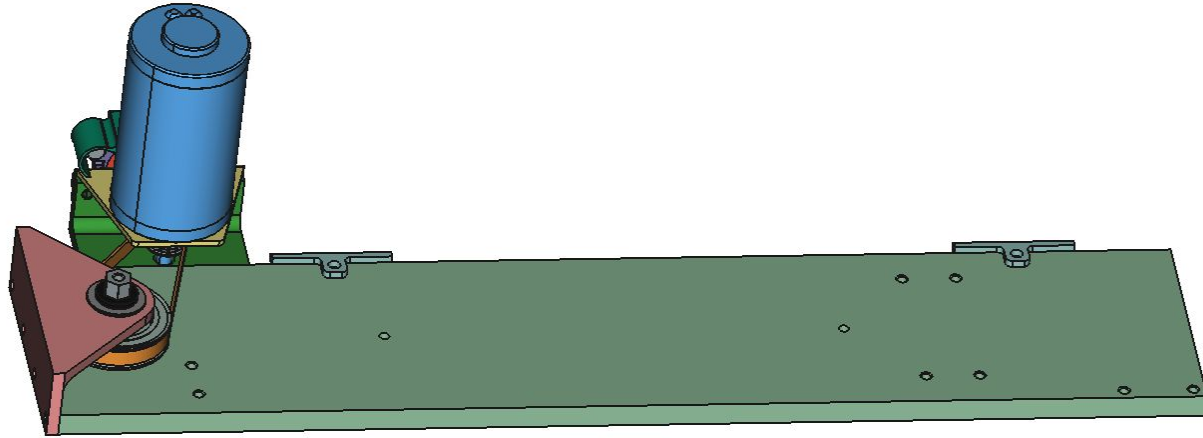
Part number:
AMAZON-B01NBOCXHE



Flat washer

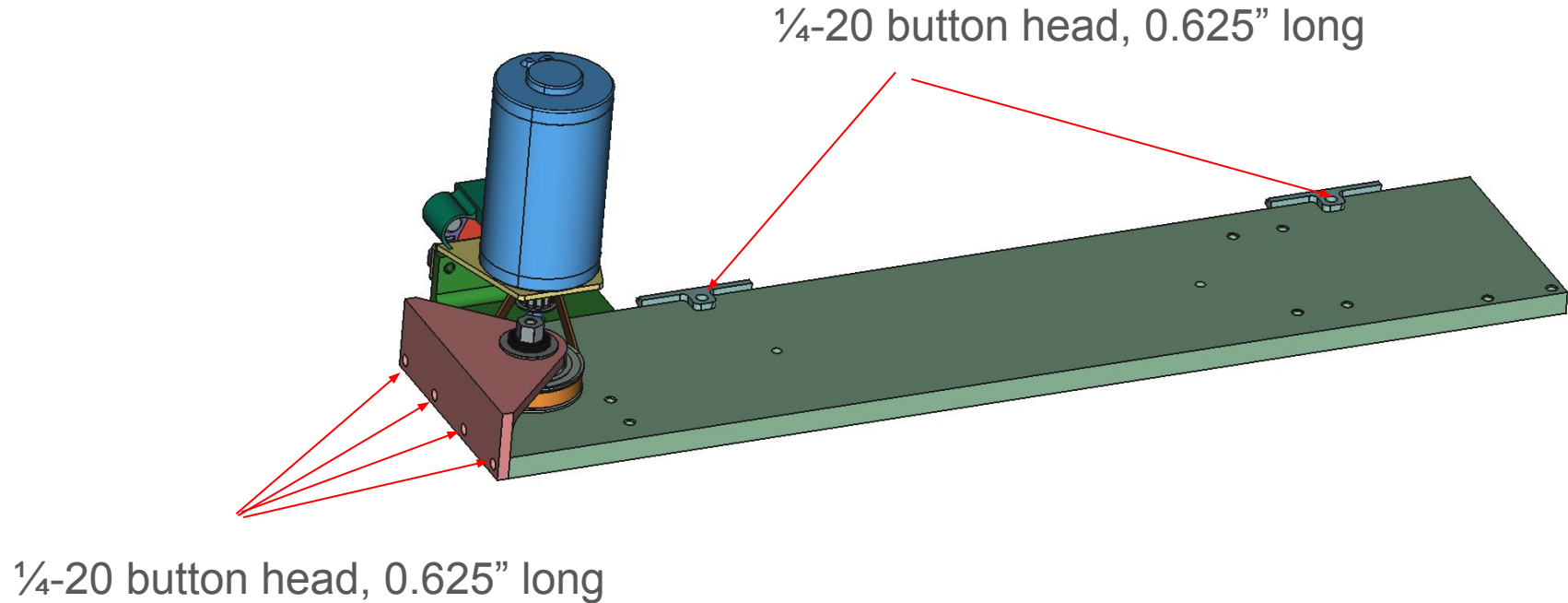
Part number:
MCMaster-90107A029

**Small quantity of other fasteners are used in specific locations called out in following instructions*

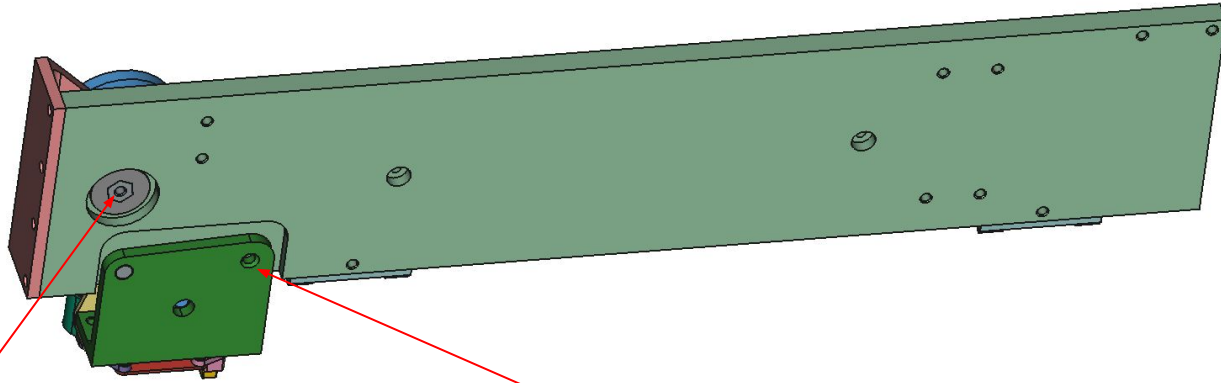


SDOF-A01 | Drive Base

SDOF-A01 | Crankshaft bracket and cable organizers



SDOF-A01 | Crankshaft bottom and belt tension



$\frac{1}{4}$ -20 button head, 0.625" long + flat washer

$\frac{1}{4}$ -20 button head

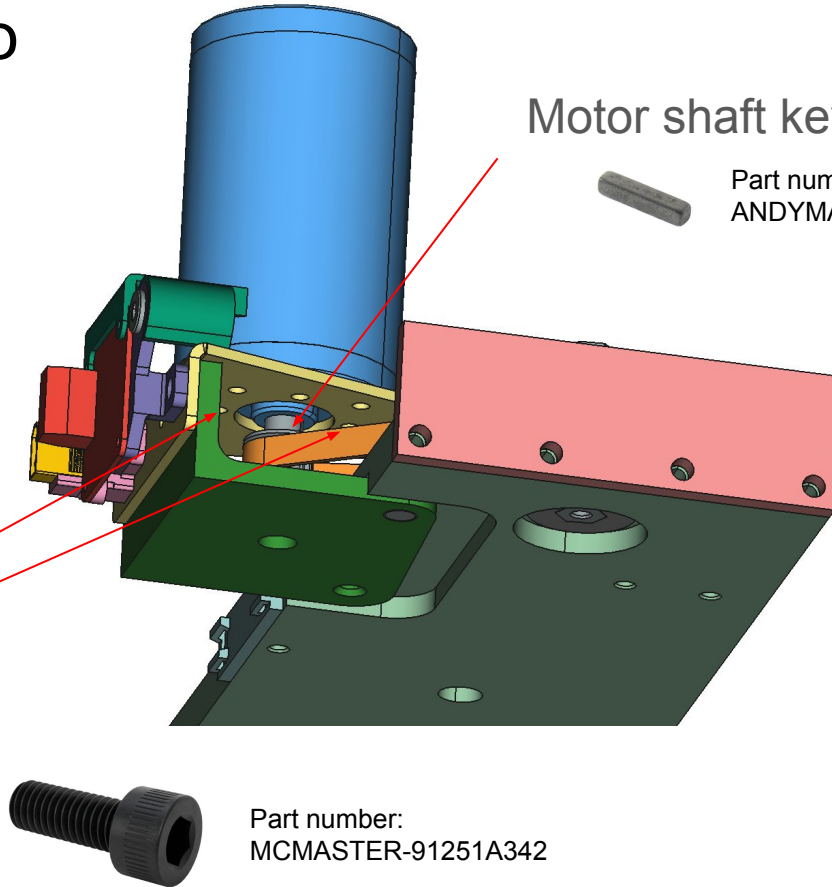
SDOF-A01 | Motor to motor bracket

Motor shaft key to pulley

Part number:
ANDYMARK-AM-1121

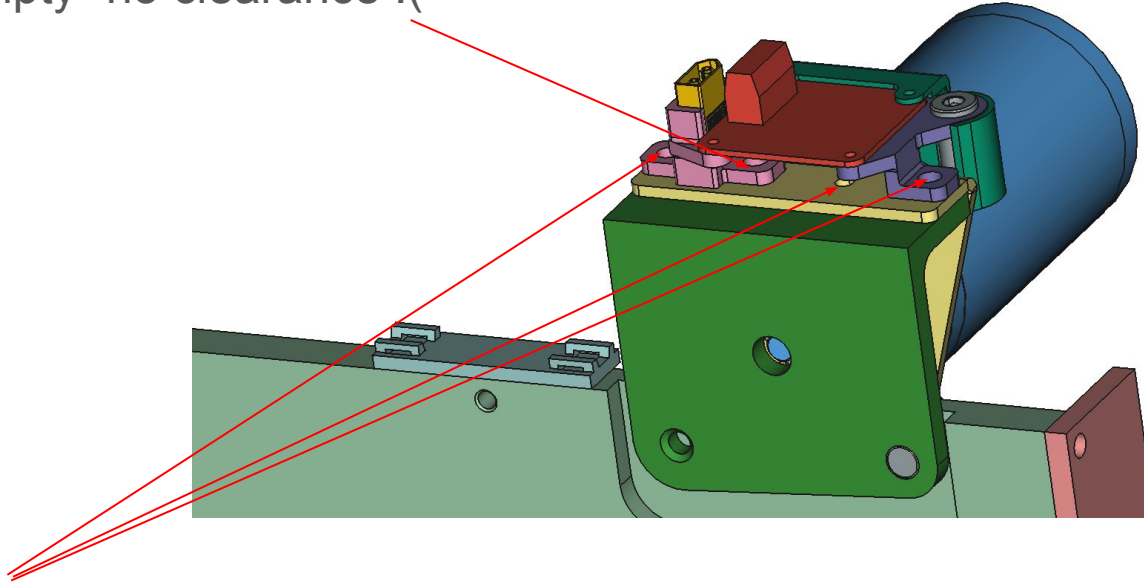
10-32 socket cap, 0.5" long

Part number:
MCMaster-91251A342



SDOF-A01 | 3 brackets to baseplate bracket

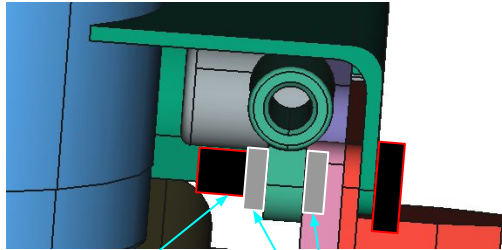
Leave empty- no clearance :(



$\frac{1}{4}$ -20 button head, 0.625" long

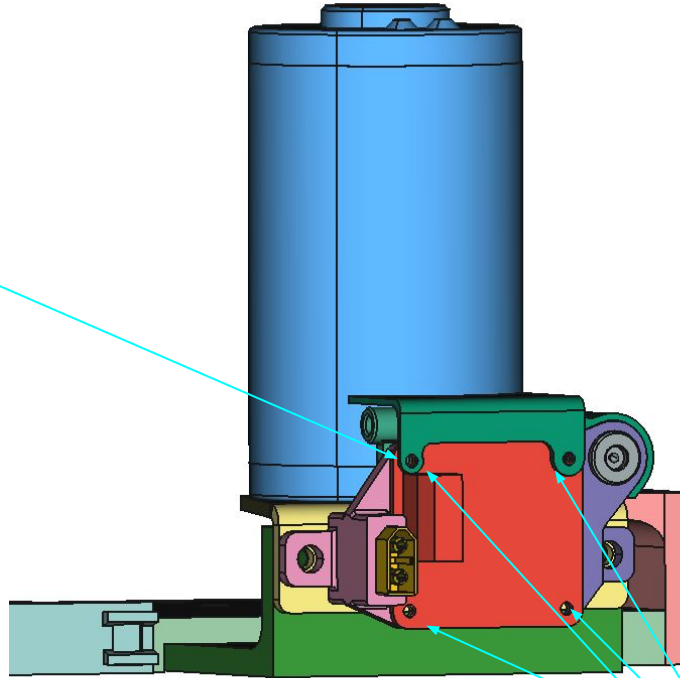
SDOF-A01 | H-bridge, covers and cable inlet

Note- use a long M3 bolt here. One nut sandwiches H-bridge to the bracket, then put on the inlet (SDOF-C09), then another nut to hold on the inlet (see below)

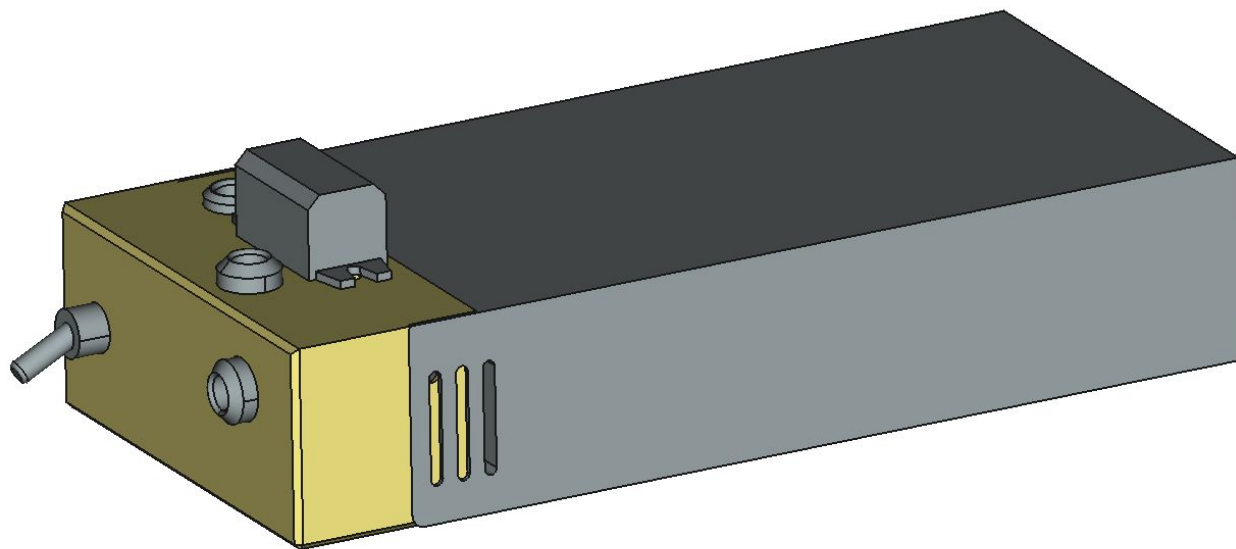


Long bolt

2x nuts

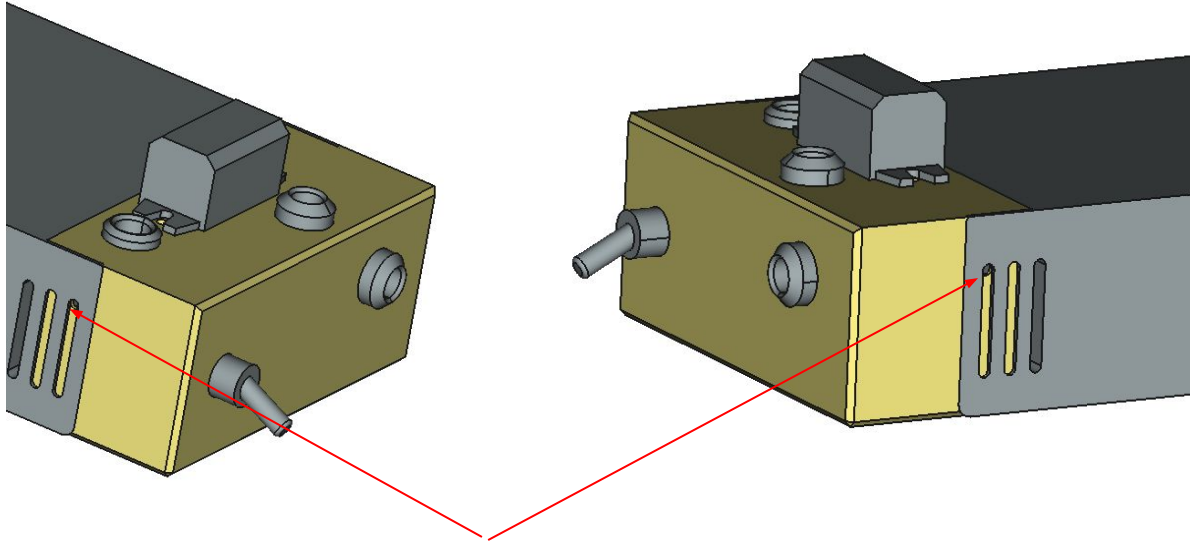


M3 machine
screws + nuts



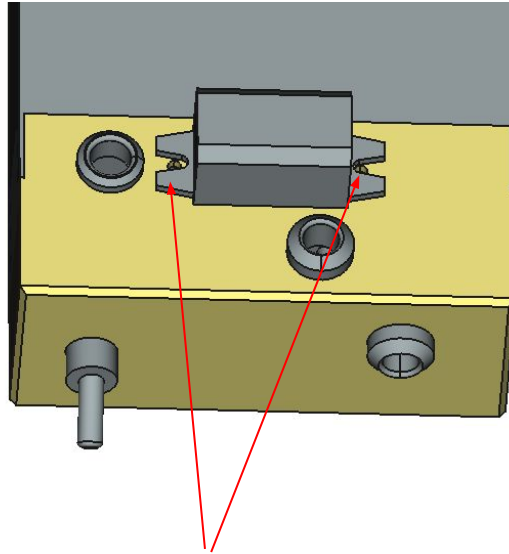
SDOF-PWR-A01 | Power Module

SDOF-PWR-A01 | Cover to power supply

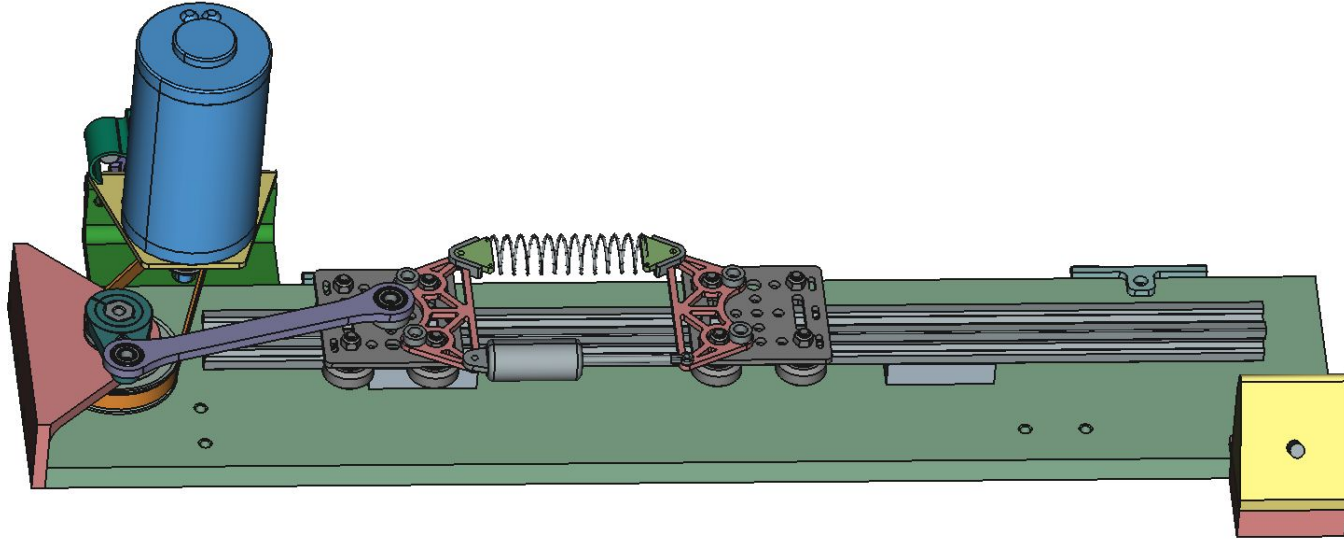


M3 machine screws + nuts

SDOF-PWR-A01 | DC-DC converter to cover

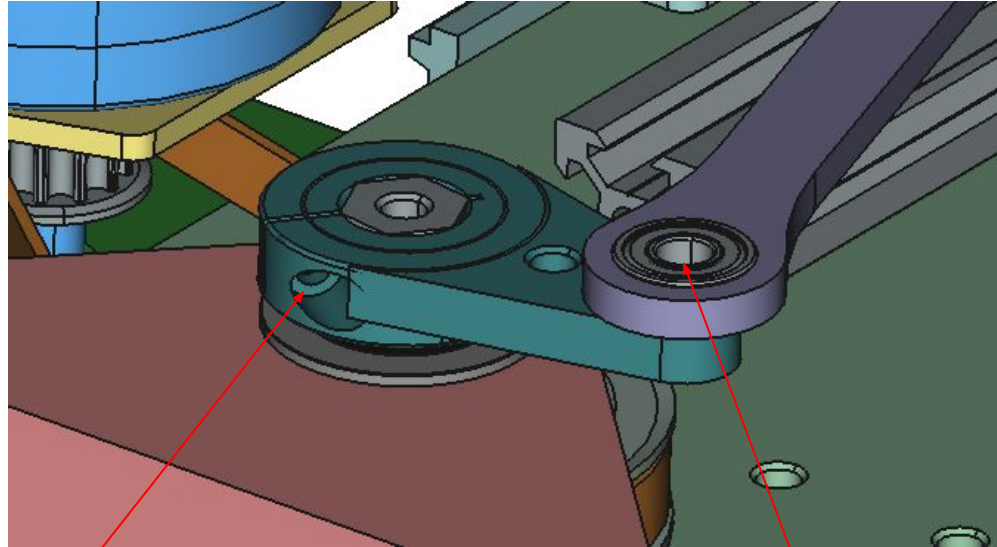


M3 machine screws + nuts



SDOF-FREQ-A01 | Frequency Response Demo

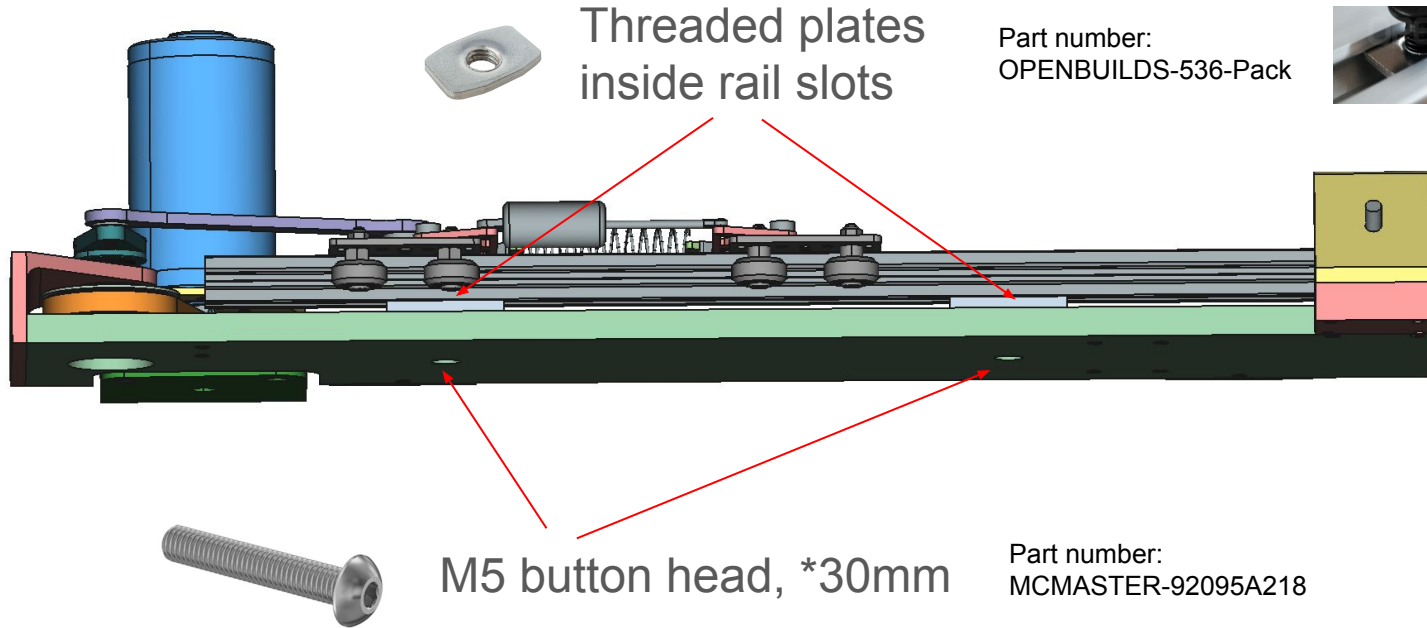
SDOF-FREQ-A01 | Crank



Pre-existing fastener ($9/64''$ allen key) to tighten crank shaft-collar

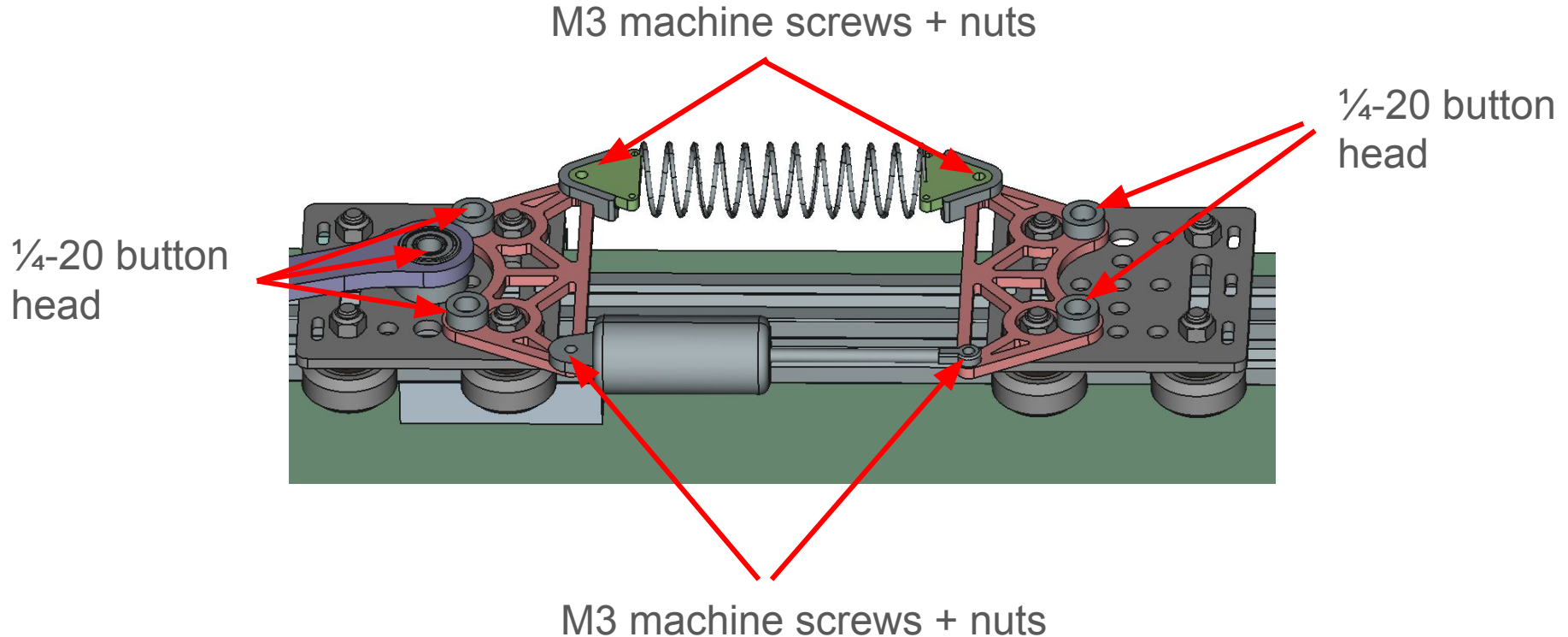
$1/4$ -20 button head

SDOF-FREQ-A01 | Linear rail to baseplate

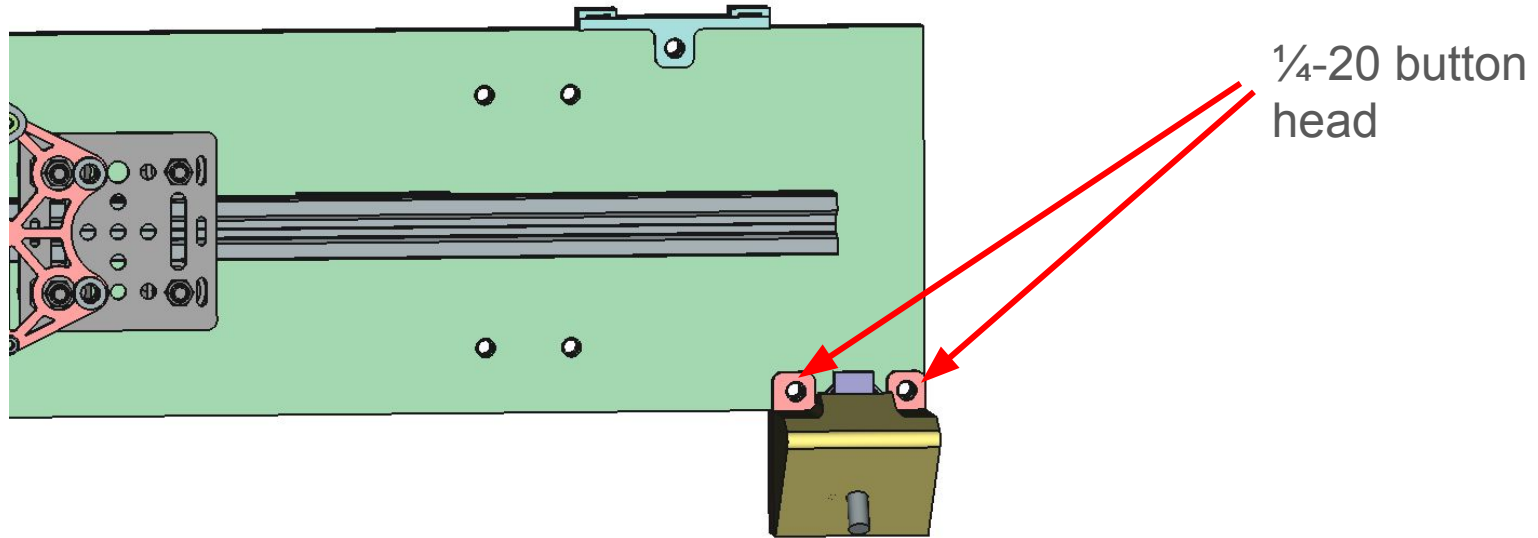


**May need to cut this a bit shorter by a couple of mm to avoid bottoming out*

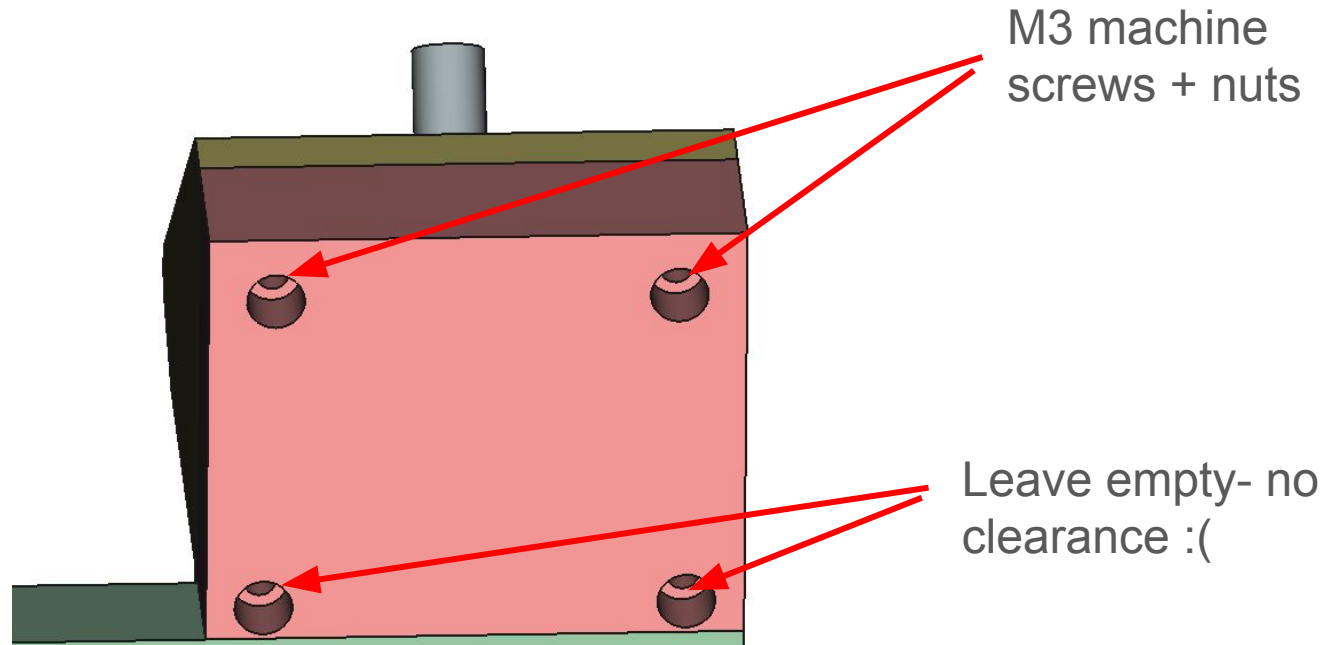
SDOF-FREQ-A01 | Mass-spring-damper assembly

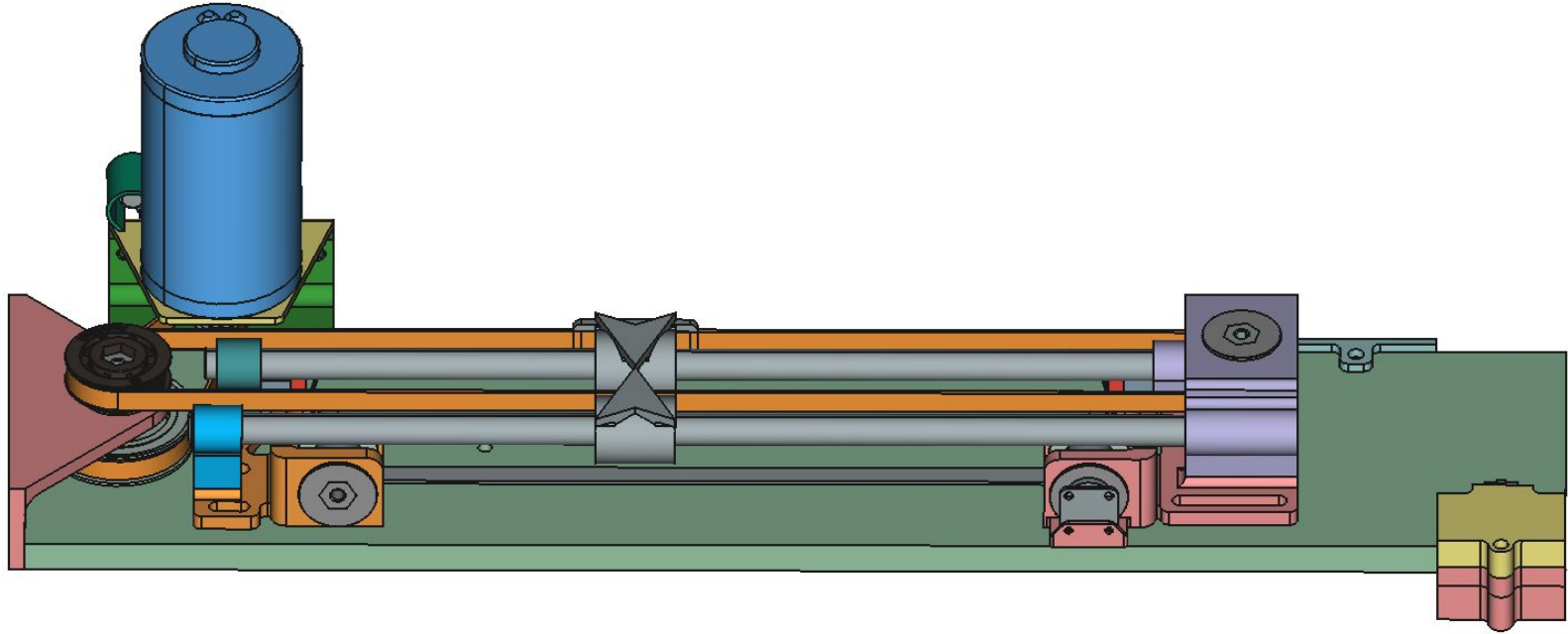


SDOF-FREQ-A01 | Frequency module to baseplate



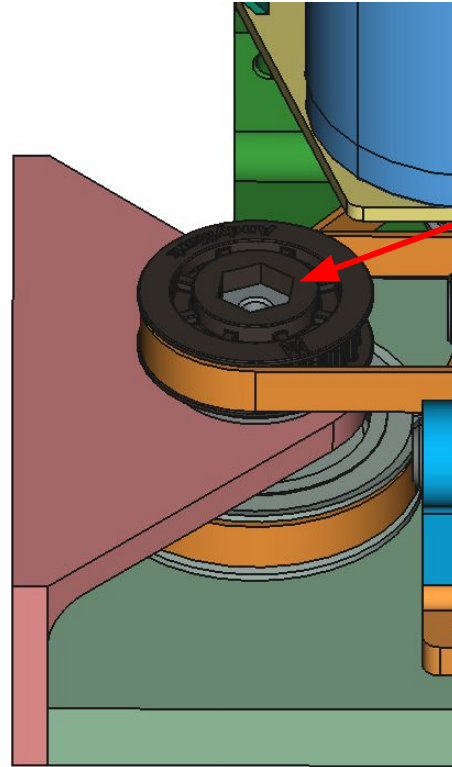
SDOF-FREQ-A02 | Frequency module lid to base





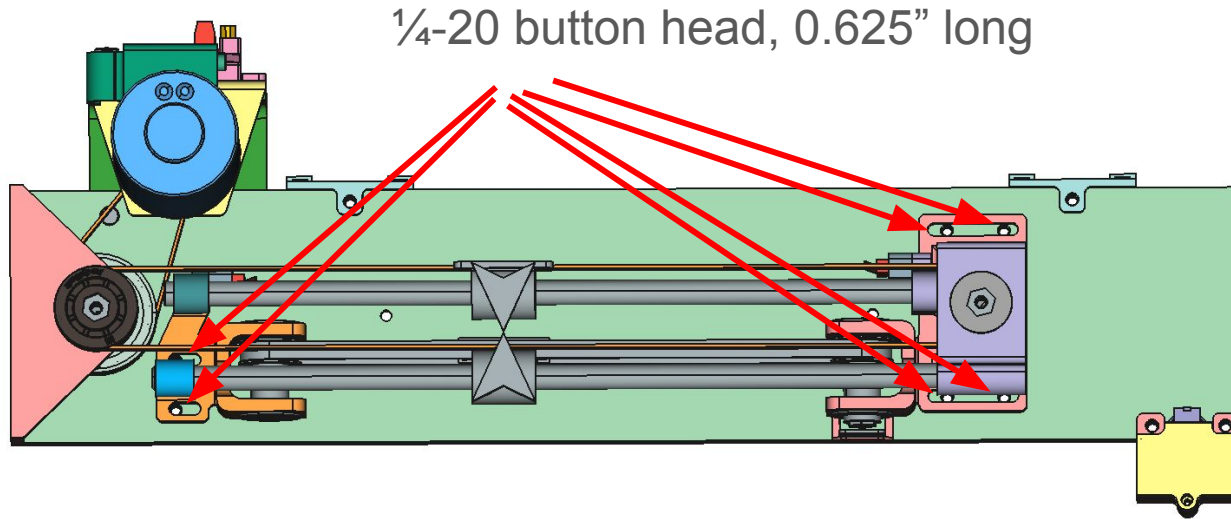
SDOF-CTRL-A01 | PID Controller Demo

SDOF-CTRL-A01 | Drive pulley to crankshaft

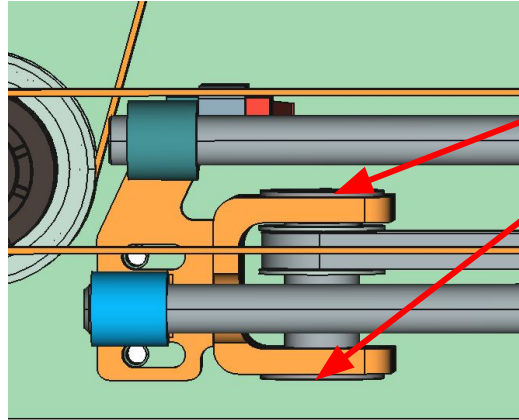


1/4-20 button head,
0.625" long + flat
washer

SDOF-CTRL-A01 | Motion assembly to baseplate



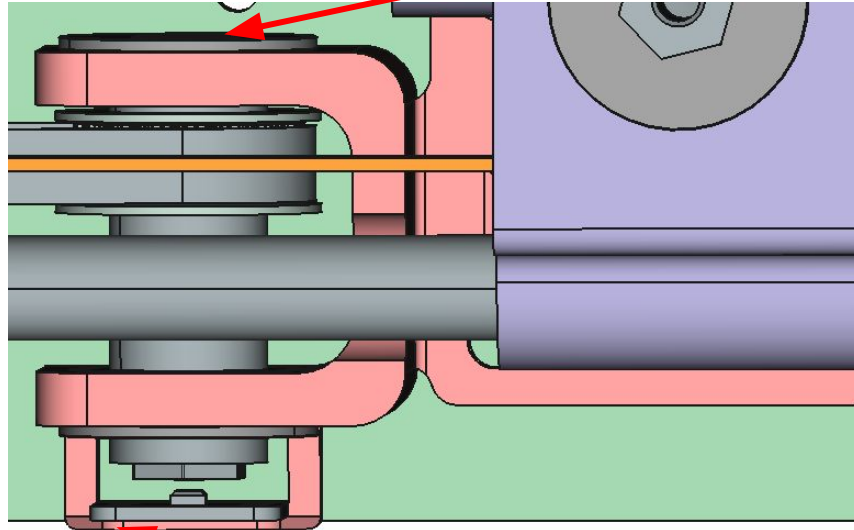
SDOF-CTRL-A01 | Left input shaft



$\frac{1}{4}$ -20 button head,
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washer

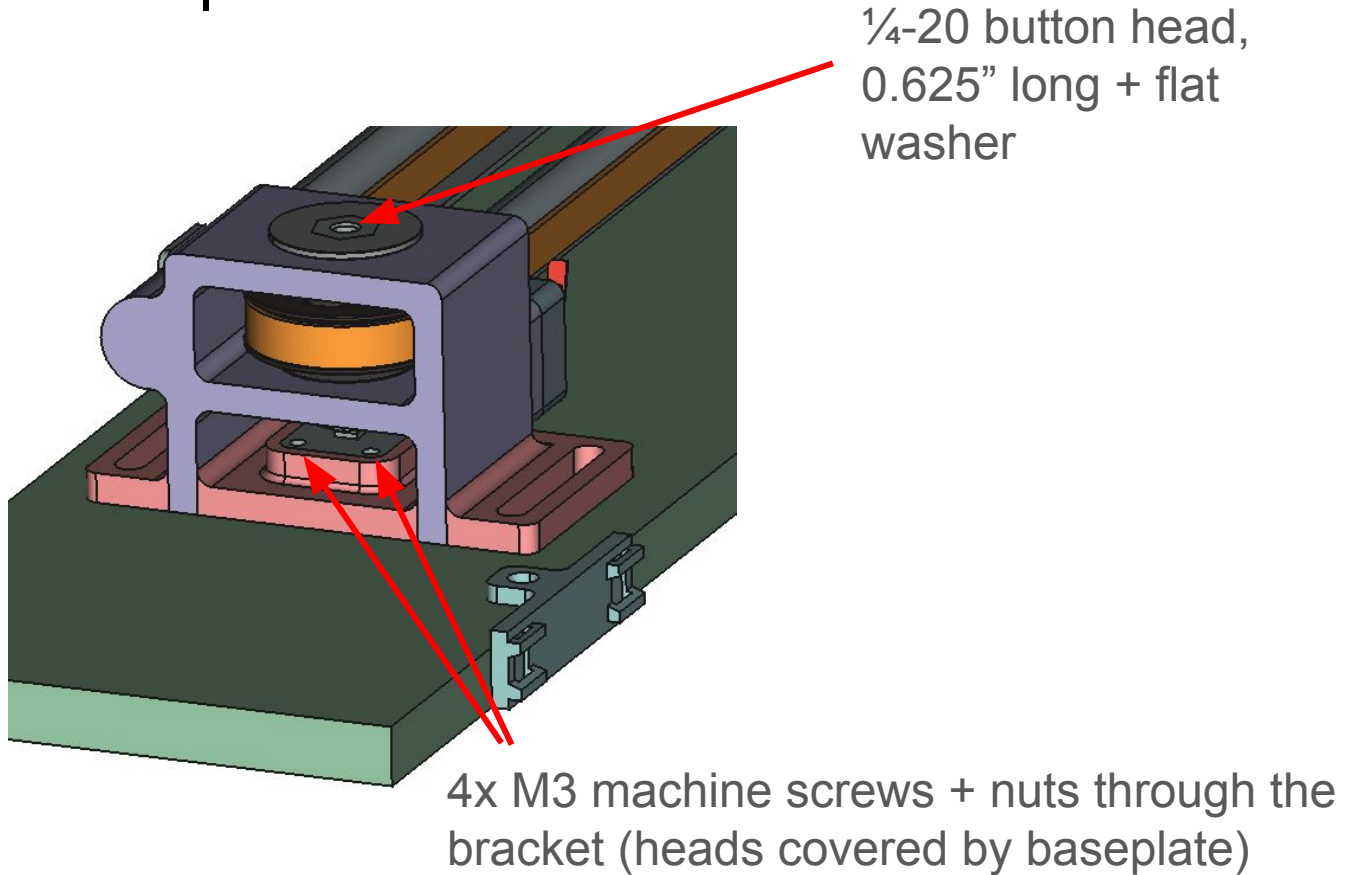
SDOF-CTRL-A01 | Right input shaft

$\frac{1}{4}$ -20 button head,
0.625" long + flat
washer

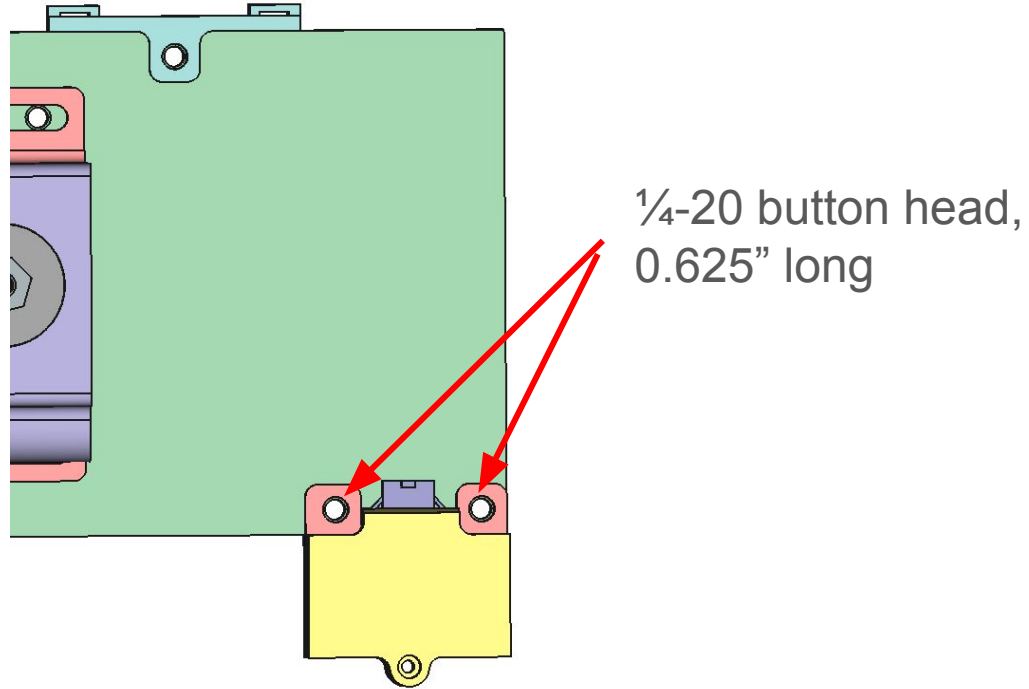


M3 machine screws + nuts

SDOF-CTRL-A01 | Driven shaft



SDOF-CTRL-A01 | Control module to baseplate



SDOF-CTRL-A02 | Control module lid to base

